



Conformational Control of a Metallo-Supramolecular Cage via the Dissymmetrical Modulation of Ligands

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Abstract: In nature as well as life systems, the presence of asymmetrical and dissymmetrical structures with specific functions is extremely common. However, the construction of metallo-supramolecular assemblies based on dissymmetrical ligands still remains a considerable challenge for avoiding the generation of unexpected isomers with similar thermodynamic stabilities, especially for three-dimensional supramolecular structures. In this study, a strategy for the conformational control of metallo-supramolecular cages via the enhancement of ligand dissymmetry was proposed. Four dissymmetrical ditopic ligands were designed and synthesized. By increasing the dissymmetry of length or angle, conformations of assemblies were precisely controlled to form discrete *cis*-Pd_nL_{2n} molecular cages.

Nature has created quantitative complicated biological systems with precise structures and function by a powerful bottom-up self-assembly approach.^[1] To mimic the fascinating self-assembly behavior in nature, supramolecular chemists have conducted considerable research by exploiting non-covalent interactions, e.g., hydrogen bonding, metal-ligand interactions, π - π stacking, dipole-dipole interactions, etc.^[2] Among these interactions, metal-ligand interactions, which exhibit high directionality and apposite bonding strength, have received considerable attention especially in 3D metal-organic cages (MOCs);^[3] which have been widely exploited in biomedicine, catalysis, molecular sensing, gas absorption, etc.,^[4] due to the presence of a large cavity and functional ligands. For the construction of these 3D structures, the most commonly employed strategy is to design highly symmetrical ligands because the high predictability and controllability of symmetrical ligands can effectively avoid the generation of unexpected isomers in the self-assembly process.

In nature, the presence of asymmetry and dissymmetry is a typical phenomenon. For example, the dissymmetry of

proteins in organisms endows rich and diverse life systems.^[5] Moreover, the dissymmetry of organic molecules is the driving force to construct chiral structures and functions.^[6] However, changing symmetry of the ligand considerably affects the structure and properties of the final assembly,^[7] and the use of dissymmetrical ligands can better simulate self-assembly in nature.^[8] However, in supramolecular self-assembly, only very few 3D metallo-supramolecular cages based on dissymmetrical ligands were obtained. There are two main reasons: firstly, the design of dissymmetrical ligands will increase the complexity of the synthetic process. Moreover, a significant challenge is how to enhance conformational control to obtain precise assembly behavior and prevent a series of unexpected isomers with similar thermodynamic stability.^[9]

The Pd_nL_{2n}-assembled cages based on the assembly of Pd(II) and bidentate organic ligands are the classic research field in MOCs.^[3e,10] For instance, Fujita et al. have successfully constructed a series of Pd_nL_{2n} cages using different angles of pyridyl ligands and Pd(II), in which “n” could be 6,^[11] 12,^[12] 24,^[13] 30,^[14] and even 48.^[15] Clever, Crowley, and others have reported a number of beneficial studies for Pd₂L₄ cages,^[16] especially the conformation of the heteroleptic cages with different symmetrical ligands can be controlled by steric and geometric constraints.^[17] However, only a few studies of Pd_nL_{2n}-assembled cages based on dissymmetrical ligands have been reported.^[18] Therefore, it is considerably valuable to do research about single-configuration self-assembly by dissymmetrical ligands.

In this study, four dissymmetrical ligands of **L1**, **L2**, **L3**, and **L4** were designed by using simple pyridine and benzene as building blocks, which were efficiently synthesized and coordinated with Pd(II) to form assembled cages Pd₂(**L1**)₄, Pd₂(**L2**)₄, Pd₂(**L3**)₄, and Pd₃(**L4**)₆ respectively (Figure 1). When the ligand **L1** was assembled with Pd(II), multiple isomers of Pd₂(**L1**)₄ were formed in the assembly process due to insufficient geometric constraints (Figure 1a).^[18a] On the basis of Pd₂(**L1**)₄, a single-configuration assembly cage is expected to be formed via the enhancement in the geometric constraints through the increase in ligand dissymmetry. All of the characterization methods (i.e., ESI-MS, NMR, single-crystal analysis, and DFT calculations) have confirmed that a single *cis*-configuration assembled cage is formed by the high dissymmetry of length (**L2** and **L3**) or angle (**L4**), which exhibits more suitable geometric constraints (Figures 1b,c). Notably, ligand **L4** can control the assembly behavior, and only a single configuration from the nine possible isomers can be obtained via the introduction of suitable geometric constraints (Figures 1c and S38).

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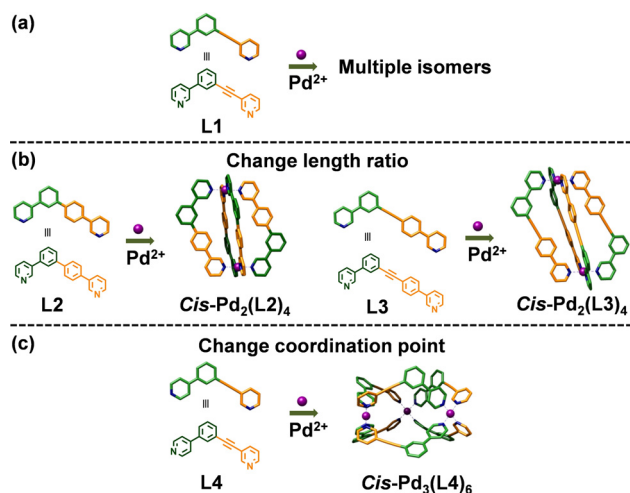


Figure 1. Self-assembly of a) $\text{Pd}_2(\text{L1})_4$, b) $\text{Pd}_2(\text{L2})_4$ and $\text{Pd}_2(\text{L3})_4$, and c) $\text{Pd}_3(\text{L4})_6$.

The synthesis and characterization of dissymmetrical ligands **L1**, **L2**, and **L3** are shown in the supporting information (Schemes S1 and S2, Figures S1–S3, S8–S10 and S23–S24). The $\text{Pd}_2(\text{L1})_4$ complex was prepared according to the methods reported previously,^[18a] and at least two isomers existed in this assembly process (Figure 1a). To further control the self-assembly process, increasing length ratio of the dissymmetrical ligand was utilized to enhance dissymmetry. The dissymmetrical ligand **L2** was designed, the molecular unilateral length of which was increased compared with **L1**. **L2** and $\text{Pd}(\text{CH}_3\text{CN})_4(\text{BF}_4)_2$ were mixed with an exact stoichiometric ratio of 2:1 in $\text{DMSO}-d_6$ to construct cage $\text{Pd}_2(\text{L2})_4$. Electrospray ionization-mass spectrometry (ESI-MS) was employed to determine the molecular composition of complex $\text{Pd}_2(\text{L2})_4$. After deconvolution, the molecular weight of complex $\text{Pd}_2(\text{L2})_4$ was 1792.3 Da, which was in good agreement with its expected chemical composition of four **L2** ligands, two Pd(II) ions, and four BF_4^- (Figures S26 and S27). In the further NMR characterization, only one set of the main product peaks was observed (Figure 2b), suggesting that increased dissymmetry can improve conformational control. However, notably, some weak signals of the unexpected isomers and by-products were still observed in the NMR spectrum (Figure 2b), even if the assembly was repeated several times, suggesting that the length ratio of ligand **L2** cannot provide completely adequate geometric constraints during the assembly process.

To further precisely control the assembly behavior of the dissymmetrical ligand and obtain a single-configuration assembly, ligand **L3** was designed by increasing again the unilateral length of the dissymmetrical ligand. By using the same assembly method, the cage $\text{Pd}_2(\text{L3})_4$ was successfully obtained. ESI-MS also confirmed that $\text{Pd}_2(\text{L3})_4$ (1888.3 Da) exhibited the same chemical composition with complex $\text{Pd}_2(\text{L2})_4$ (Figures S28 and S29). Compared with the ^1H NMR of cage $\text{Pd}_2(\text{L2})_4$, there is only a set of peaks in ^1H NMR spectrum of cage $\text{Pd}_2(\text{L3})_4$ (Figure 2c), indicating that enhancing the ligand dissymmetry by further increasing the length ratio can raise the geometric constraints of the

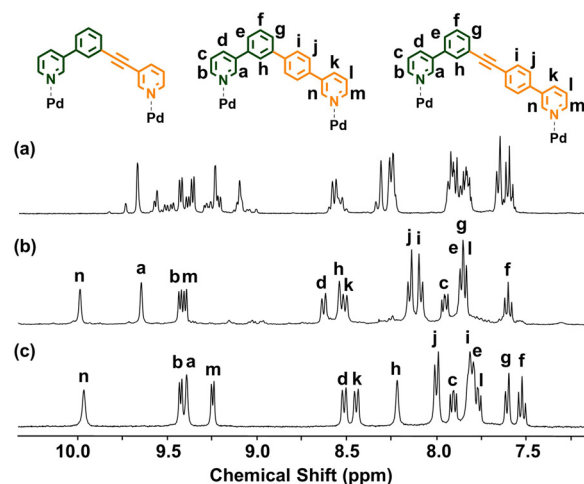


Figure 2. ^1H NMR spectra (400 MHz, 300 K) of (a) $\text{Pd}_2(\text{L1})_4$, (b) $\text{Pd}_2(\text{L2})_4$, and (c) $\text{Pd}_2(\text{L3})_4$ in $[\text{D}_6]\text{DMSO}$.

assembly to control the self-assembly behaviors. In addition, the proton signals of $\text{Pd}_2(\text{L2})_4$ and $\text{Pd}_2(\text{L3})_4$ were assigned by 2D COSY NMR (Figures S6 and S13) and 2D NOESY (Figures S7 and S14). The observed cross-peaks ($\text{H}_a \cdots \text{H}_n$ and $\text{H}_b \cdots \text{H}_m$, Figures S7 and S14) illustrate the intramolecular through-space interaction of two pyridine in ligands **L2** and **L3**, suggesting *cis* or *trans* isomers should be formed in self-assembly process.

However, NMR and mass spectrometry cannot fully prove the specific configuration of $\text{Pd}_2(\text{L2})_4$ and $\text{Pd}_2(\text{L3})_4$ due to the possible presence of isomers. To further investigate details of supramolecular structures, X-ray diffraction was utilized to characterize the specific structural configuration of the two assemblies. Single crystals of $\text{Pd}_2(\text{L2})_4$ and $\text{Pd}_2(\text{L3})_4$ were obtained by slow diffusion of THF vapor into CH_3CN containing $\text{Pd}_2(\text{L2})_4$ or DMSO containing $\text{Pd}_2(\text{L3})_4$ for 3 weeks and were tested by synchrotron radiation X-ray diffraction. X-ray diffraction data indicated that $\text{Pd}_2(\text{L2})_4$ and $\text{Pd}_2(\text{L3})_4$ belong to the triclinic system, with the $P\bar{1}$ space group, and the detailed crystal information was shown in the Supporting Information Figures S32–S35 and Tables S1–S2. The crystal structure of $\text{Pd}_2(\text{L2})_4$ exhibited a *cis*-configuration, in which the long side and the short side of the ligands were complementarily connected. The two ligands on one plane were obviously distorted (Figure 3a), while the two ligands on the other plane were almost undistorted (Figure 3b), further indicating that the dissymmetry of ligand **L2** is not sufficient to bring geometric constraints in the assembly process. Figures 3c and 3d showed the single-crystal structure of $\text{Pd}_2(\text{L3})_4$. To our surprise, two structures of $\text{Pd}_2(\text{L3})_4$, defined as $\text{Pd}_2(\text{L3})_4\text{-A}$ and $\text{Pd}_2(\text{L3})_4\text{-B}$, respectively, were observed, and both structures exhibited a highly similar *cis*-configuration. For $\text{Pd}_2(\text{L3})_4\text{-A}$, the acetylene bonds of the four ligands were only slightly twisted, showing a stable conformation (Figures 3e,f). However, for $\text{Pd}_2(\text{L3})_4\text{-B}$, the acetylenic bonds of the two ligands on a plane were almost not bent (Figure 3g), but the acetylene bonds of the two ligands on the other plane were clearly bent in the opposite direction (Figure 3h). Low temperature ^1H NMR in $\text{DMF}-d_7$

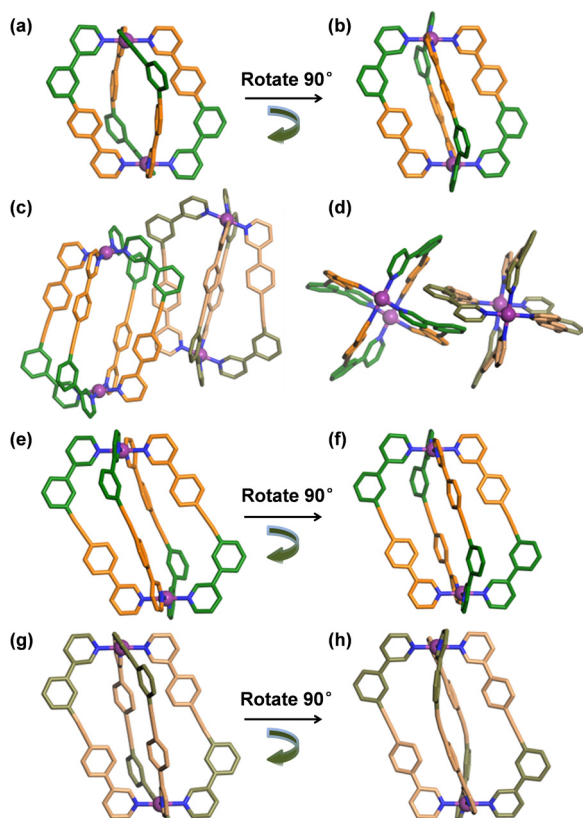


Figure 3. Single-crystal structures of $\text{Pd}_2(\text{L}2)_4$ (a) and (b), $\text{Pd}_2(\text{L}3)_4$ (c) and (d), $\text{Pd}_2(\text{L}3)_4\text{-A}$ (e) and (f), and $\text{Pd}_2(\text{L}3)_4\text{-B}$ (g) and (h). Views of (b), (f), and (h) are rotated 90° clockwise along the Pd-Pd axis from (a), (e), and (g), and hydrogen atoms are omitted for clarity.^[19]

was tested and showed just one set of broadened peaks, suggesting only one isomer in solution state (Figure S15). The two conformers of $\text{Pd}_2(\text{L}3)_4$ obtained from crystal data should be attributed to intermolecular packing and the moderate flexibility of acetylene bond. It also revealed that the geometric constraints originating from the dissymmetry of **L3** can efficiently control the existence of *cis*-configuration without other isomers. By the analysis of the crystal structures of $\text{Pd}_2(\text{L}2)_4$ and $\text{Pd}_2(\text{L}3)_4$, the self-assembly process can be completely confirmed to be controlled via the increase in ligand dissymmetry.

To further understand the effect of ligand dissymmetry on the assembly results, density functional theory (DFT) calculations were employed to calculate the energy of the isomers (Figure 4). To be consistent with the experiment, the effects of solvent were considered in the DMSO solution using the polarizable continuum model (PCM). For $[\text{Pd}_2(\text{L}1)_4]^{4+}$, the energy difference between two isomers is 0.1 kJ mol^{-1} , which was confirmed by the complicated NMR spectrum and DFT simulations in the Jelfs' work.^[18a] For $[\text{Pd}_2(\text{L}2)_4]^{4+}$, the calculated relative energies of four isomers $[\text{Pd}_2(\text{L}2)_4]^{4+}$ are in the order of $\text{C} < \text{D} < \text{B} < \text{A}$ (isomer C exhibited the *cis*-configuration assembly). To verify the effect of solvent on the relative energy of $[\text{Pd}_2(\text{L}2)_4]^{4+}$, DFT calculations were also performed in MeCN solution. The calculated results indicated that the solvent has no significant effect in the relative energies of the four isomers (Table S4). The structure of the

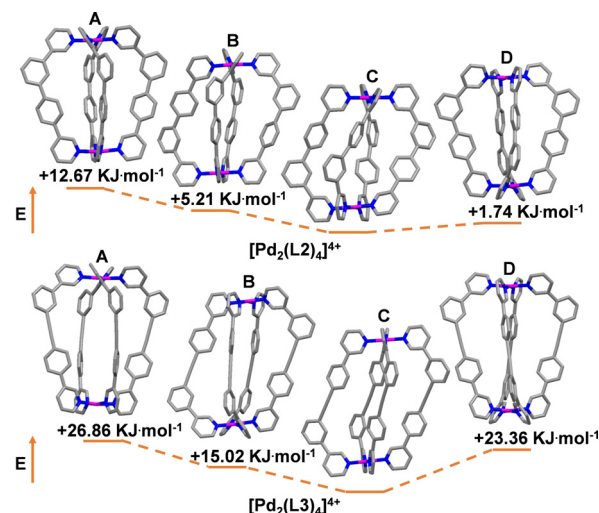


Figure 4. Calculated structures (B3LYP/6-31G(d)/SDD) of the isomers of $[\text{Pd}_2(\text{L}2)_4]^{4+}$ and $[\text{Pd}_2(\text{L}3)_4]^{4+}$ with the energies relative to isomer C (Hydrogen atoms are omitted for clarity).

isomer C was consistent with the available crystal structures of $[\text{Pd}_2(\text{L}2)_4]^{4+}$. The energy difference between the isomers C and D was 1.74 kJ mol^{-1} , and the ^1H NMR spectrum also revealed weak signals of the unexpected isomers, which indicated that there may exist the isomers C and D at the room temperature ($< 2.51 \text{ kJ mol}^{-1}$). The calculated relative energies of the isomers of $[\text{Pd}_2(\text{L}3)_4]^{4+}$ are in the order of $\text{C} < \text{B} < \text{D} < \text{A}$. Particularly, the energy difference between the isomers B and C was extraordinarily high ($15.02 \text{ kJ mol}^{-1}$), which was energetically favorable for the formation of a uniform *cis*-configuration $[\text{Pd}_2(\text{L}3)_4]^{4+}$ cage with high purity. These DFT simulations were consistent with the stable structure obtained by experiment.

Encouraged by the successful control of the self-assembly process by changing the dissymmetry of two arms, we tried to change the angle of the coordination position of **L1** to increase the ligand dissymmetry and to study whether it can bring appropriate geometric constraints in the assembly to obtain a single-configuration assembly. Therefore, **L4** was designed and synthesized, which is an isomer of **L1**. The synthesis and characterization of **L4** were described in the supporting information (Scheme S3, Figures S16–S18 and S25). The structure of **L4** comprised pyridine at the para position on one side and pyridine at the meta position on the other side. Ligand **L4** and $\text{Pd}(\text{CH}_3\text{CN})_4(\text{BF}_4)_2$ were dissolved in 1.0 mL of $\text{DMSO-}d_6$ and heated at 70°C for 8 h . The assembly formed by the coordination of **L4** and $\text{Pd}(\text{II})$ was speculated to form a three-metal hexamer structure, which may comprise nine isomers (Figure S38). However, ^1H NMR (Figure 5a) and ^{13}C NMR (Figure S20) spectra of $\text{Pd}_3(\text{L}4)_6$ clearly revealed only one set of product peaks, indicating the formation of a single product. All proton signals were assigned by 2D COSY and 2D NOESY NMR (Figures S21 and S22). Moreover, the ESI-MS spectrum also exhibited a series of signal peaks at $m/z = 1101.1$, 705.1 , and 507.1 , corresponding to the charge states $2+$, $3+$, and $4+$, resulting from the successive loss of the counterions BF_4^- . Each isotope pattern of these peaks matched well with the

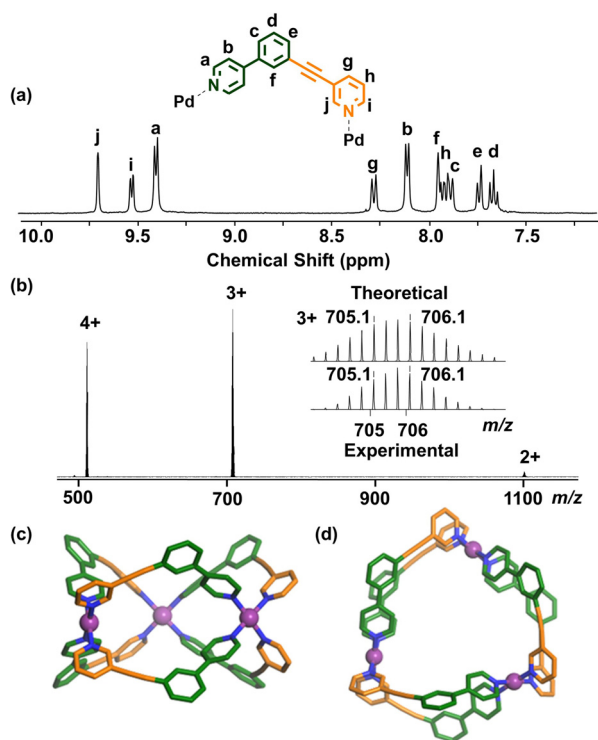


Figure 5. (a) ^1H NMR spectrum (400 MHz, 300 K), (b) ESI-MS, (c) the front view, and (d) the top view of the crystal structure of $\text{Pd}_3(\text{L4})_6$ (Hydrogen atoms are omitted for clarity).^[19]

corresponding simulated isotope pattern of $\text{Pd}_3(\text{L4})_6$ with a molecular weight of 2376.33 Da (Figures 5b, S30, and S31). This confirmed that the assembly $\text{Pd}_3(\text{L4})_6$ is a hexamer structure composed of six ligands, three Pd^{2+} , and six BF_4^- . X-ray crystal diffraction was utilized to characterize the specific configuration of $\text{Pd}_3(\text{L4})_6$. The poor solvent THF was slowly diffused into MeCN solution of $\text{Pd}_3(\text{L4})_6$. After about 3 weeks, the crystal of supramolecule $\text{Pd}_3(\text{L4})_6$ was obtained. X-ray diffraction data indicated that $\text{Pd}_3(\text{L4})_6$ belongs to the triclinic system, with the $P\bar{1}$ space group, and the detailed crystal information of $\text{Pd}_3(\text{L4})_6$ was in the supporting information (Figures S36–S37 and Table S3). The single crystal structure of $\text{Pd}_3(\text{L4})_6$ was shown in Figure 5c, which has proved that $\text{Pd}_3(\text{L4})_6$ still exhibited a single *cis*-configuration with the long side and the short side of ligands complementary to connect, and each Pd(II) was coordinated by two pyridines at the meta position and two pyridines at the para position. The optimized geometrical structures and calculated relative energies of nine isomers were shown in Figure S39. The *cis*- $\text{Pd}_3(\text{L4})_6$ is the most energetically favorable structure among nine isomers. Therefore, the calculated results confirmed that *cis*- $\text{Pd}_3(\text{L4})_6$ is the most stable structure, which is well consistent with the obtained crystal structure of *cis*- $\text{Pd}_3(\text{L4})_6$. This indicated that a reasonable design of the coordination site position of dissymmetrical ligands can render appropriate geometric constraints in the assembly to form an assembly of a single configuration.

In summary, four dissymmetrical ligands **L1**–**L4** were synthesized. Compared with ligand **L1**, the length ratios of **L2** and **L3** were gradually increased, and the coordination

position on one side of **L4** was changed from meta to para. After coordination with Pd(II), the obtained complexes were fully characterized through analysis of ESI-MS, NMR, single-crystal data, and DFT calculations. Compared with different isomers based on ligand **L1**, the self-assembly of **L2**, **L3**, and **L4** with increasing dissymmetry of length or angle revealed extremely good conformational control, and especially in the system of $\text{Pd}_2(\text{L3})_4$ and $\text{Pd}_3(\text{L4})_6$, only a single *cis*-configuration existed. The results indicated that the strategy by increasing dissymmetry to realize the conformational control of dissymmetrical ligands in self-assembly is extremely efficient. This work offered a new route for designing complex supramolecular architectures with high controllability based on dissymmetrical ligands. Moreover, it will promote extensive attention and research for the scientists from different aspects to further understand the self-assembly based on dissymmetrical ligands.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: cage compounds · conformational control · dissymmetrical ligand · self-assembly · supramolecular chemistry

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